

MCU Board for the Basic Pixels Powered Zoetrope Platter

This board has a STM32 microcontroller at its heart

Power is passed to the platter via 2.54 male headers. a female connector on the goosenek slipring connects into the header.

A sensor is used to detect the rotation speed and index point of the rotating platter. The sensor is a on the other PCB and single connection is passed to this board

There are two, five pin connections on two ends of this PCB. These are connections to the two neopixel strip boards that act as the display. On one of these boards an optical sensor is mounted for rotation speed sensing and indexing of the platter

There is a single SMT LED on this PCB to act as a heartbeat/debug LED. It may be incorporated into animations if desired

This board, to start, is square. Each edge is tangent to the platter coupler

MCU



Power



Connectors

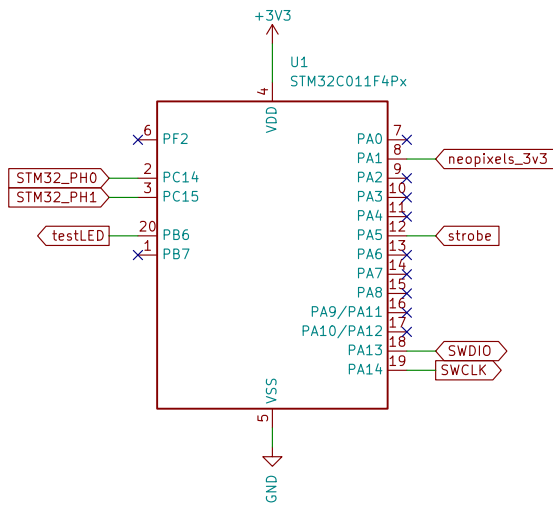


Level Shifting

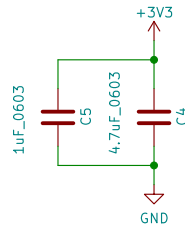


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KiCad E.D.A. 10.0.0	Id: 1/5	

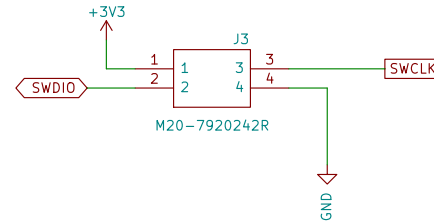
Small form factor MCU



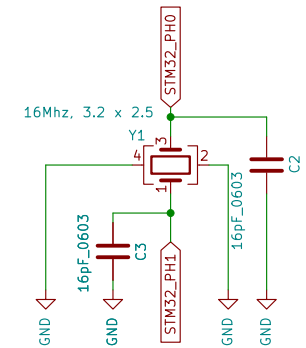
STM32 Bypass Caps



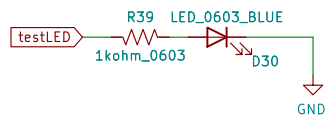
Programming Header



Main Crystal



heartbeat LED



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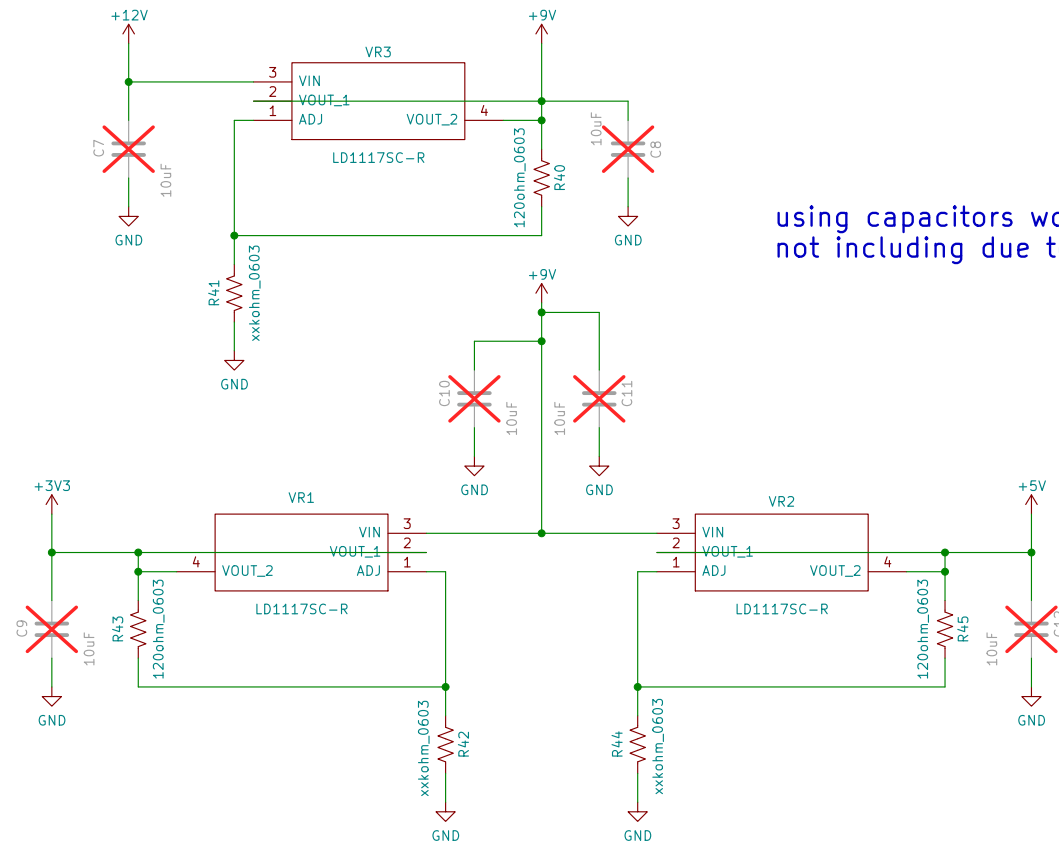
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Three LDOs for power to the Platter

9VDC LDO takes the 12V to 9V, this is here for heat reasons – bypass available in case its not necessary

5VDC LDO for the Neopixels and index/speed sensor

3VDC for MCU



using capacitors would improve ripple,
not including due to parts on hand

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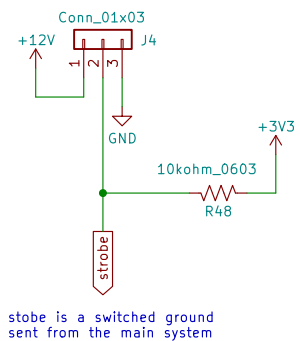
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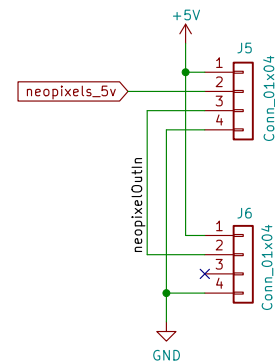
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connection from main system



connections to pixel blades



Sheet: /Connectors/
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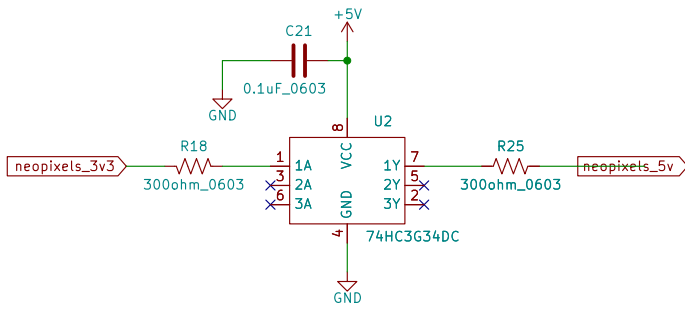
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3.3V to 5V Level Shifter



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